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Revisiting MAE pre-training for 3D medical image segmentation

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Abstract

Self-Supervised Learning (SSL) presents an exciting opportunity to unlock the potential of vast, untapped clinical datasets, for various downstream applications that suffer from the scarcity of labeled data. While SSL has revolutionized fields like natural language processing and computer vision, its adoption in 3D medical image computing has been limited by three key pitfalls: Small pre-training dataset sizes, architectures inadequate for 3D medical image analysis, and insufficient evaluation practices. In this paper, we address these issues by i) leveraging a large-scale dataset of 39k 3D brain MRI volumes and ii) using a Residual Encoder U-Net architecture within the state-of-the-art nnU-Net framework. iii) A robust development framework, incorporating 5 development and 8 testing brain MRI segmentation datasets, allowed performance-driven design decisions to optimize the simple concept of Masked Auto Encoders (MAEs) for 3D CNNs. The resulting model not only surpasses previous SSL methods but also outperforms the strong nnU-Net baseline by an average of approximately 3 Dice points setting a new state-of-the-art. Our code and models are made available here.

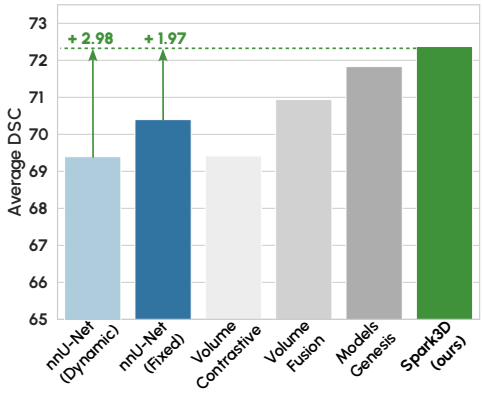

Figure 1. **Well-configured MAE pre-training for CNNs is state-of-the-art.** When comparing our MAE pre-trained model on all eleven test Datasets, Spark3D achieves almost +3 DSC points compared to the strong nnU-Net baseline and outperforms current SSL methods.

1. Introduction

In recent years, the concept of Self-Supervised Learning (SSL) has emerged as a driving factor in data-rich domains, enabling large-scale pre-training that facilitates the learning of robust and transferable general-purpose representations [2, 16, 28]. This paradigm shift has been instrumental in advancing various fields, particularly in domains with abundant labeled data like NLP or natural vision. In the domain of 3D medical image computing, this trend has not caught on. Currently, the domain is either focused on training from-scratch, mainly using the nnU-Net framework by Isensee et al. [20], or using supervised pre-training, which is limited by the cost associated with annotated data [19, 38, 44]. The usage of supervised pre-training implies the willingness of the domain to adopt pre-training and calls to question the currently established SSL methods in the domain. We believe this lack of widespread adoption of previously established methods can be attributed to three major pitfalls of previous SSL research in this field:

P1 - Limited dataset size: Many SSL approaches have been developed on very few unlabeled volumes, often being trained on fewer than 10,000 images [3, 13, 14, 33, 34,

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46, 48, 49], almost approaching scales of supervised dataset sizes. These datasets tend to be pooled from publicly available annotated datasets, as larger datasets pose a greater hurdle to acquire. While many hospitals possess 3D medical images in the millions, they are locked away from the public due to patient privacy concerns. Some larger open-source datasets exist, e.g., the Adolescent Brain Cognitive Development (ABCD) dataset of the NIH (N=40k) [27] or the UK Biobank dataset (N=120k), but they restrict access pending an internal review board's approval, posing a hurdle for open science.

P2 - Outdated Backbones: Many studies develop their SSL method on non-state-of-the-art architectures, e.g., utilizing transformers [3, 6, 13, 34, 43, 46]. While transformers are prevalent in the 2D natural imaging domain [9], recent architectures leveraging attention [40] have so far not been able to reach state-of-the-art performance in 3D medical segmentation. In fact, well-configured 3D U-Net [7, 30] inspired CNNs dominate 3D medical image segmentation, outperforming transformer-based models by a large margin [21]. This underscores the need for SSL methods that can be seamlessly integrated with CNNs to harness their full potential in medical image analysis on downstream tasks.

P3 - Insufficient Evaluation: Existing methods often lack rigorous evaluation, masking the methods' efficacy. This is represented through: i) Evaluating too few datasets to show generalization of the pre-training [3, 39] ii) Stacking multiple contributions, e.g., novel architecture with new pre-training, which does not allow one to draw conclusions if the pre-training is effective without the architecture [43] iii) Comparing to bad from-scratch baselines, like a badly configured, outdated Çiçek et al. [7] UNet instead of a well-configured 3D nnU-Net CNN baseline [20]. iv) Evaluating their method on seen data they pre-trained on. We want to emphasize that we do not intend to point fingers, but to raise awareness that evaluation matters and insufficient evaluation can lead to a lack of clarity which methods are the best. This observation is similar to the recent study by Isensee et al. [21], which shows that this is also prevalent in the field of medical image segmentation when training from scratch.

In this paper, we explore the Masked Auto Encoder (MAE) paradigm for 3D CNNs, with the recent adaptations introduced by Tian et al. [35], Woo et al. [45]. Our approach addresses the key pitfalls of previous SSL research in 3D medical image segmentation, ensuring broader applicability and higher performance. Our main contributions are as follows:

1. **State-of-the-Art Performance:** Our final model achieves superior performance compared to existing SSL-pre-trained models using the same architecture and pre-training dataset, see Fig. 1. Notably, this is the first work to demonstrate that SSL pre-training with a fixed architecture can consistently outperform a state-of-the-art, dynamically optimized nnU-Net baseline by approximately 3 Dice points across 11 diverse downstream datasets [20].
2. **Comprehensive Evaluation Addressing Previous Pitfalls:** Building on the detailed analysis of limitations in prior work (see Tab. 1), we set up a robust development environment to design our method. (i) P1: We utilize a collection of 39k 3D MRI volumes for self-supervised pre-training, substantially larger than most datasets used in prior SSL work [14, 33, 34, 39, 46, 48, 49]. (ii) P2: By using the state-of-the-art Residual Encoder U-Net CNN architecture [21], we bypass the limitations of outdated or transformer-based architectures, leveraging CNNs' superior performance in 3D medical segmentation. (iii) P3: We rigorously evaluate our model across five development and eight testing datasets, covering a wide range of downstream tasks, diverse pathologies, novel image modalities, and multi-center datasets. Moreover, we disentangle the pre-training methodology from other contributions like novel backbones and include the strong 3D nnU-Net baseline, ensuring reliable results.
3. **In-Depth Ablation Studies:** We provide detailed analyses of critical design decisions, such as masking ratio, sparsification, and fine-tuning strategies. Additional ablations explore generalization, low-data performance, and faster fine-tuning, offering valuable insights.

Table 1. **Current SSL methods are limited by three pitfalls:** They are trained on less than 10K 3D images (P1), they do not use a state-of-the-art backbone for 3D image segmentation as recommended by Isensee et al. [21] (P2) or they do not evaluate their method on more than 4 downstream datasets (P3). ✓ indicates evading a pitfall while ✗ indicates falling for a pitfall.

Method	Year	P1 Data	P2 Model	P3 Eval.
3DMAE [6]	Jan. 2023	✗	✗	✗
GL-MAE [49]	Aug. 2023	✗	✗	✓
MAPSeg [47]	March 2023	✗	✗	✗
HySparK [33]	Aug. 2024	✗	✓	✓
AMAES [25]	Aug. 2024	✓	✓	✗
Swin UNETR [34]	March 2022	✗	✗	✓
VoCo [46]	April 2024	✗	✗	✓
VF [43]	June 2023	✓	✗	✗
MG [48]	Aug. 2019	✗	✗	✓
GVSL [17]	March 2023	✗	✗	✓
S3D (ours)		✓	✓	✓

While the fundamental concept of an MAE is not novel, its application in 3D medical image analysis has yet to be

fully validated due to limitations in previous studies, which we illustrate in Tab. 1. Our carefully designed experiments close these gaps and reveal MAEs true potential.

2. Development Framework

The goal of this paper is to develop a robust SSL pre-training method. Due to the limited prior work in the 3D medical domain, many design choices need to be made. We address this by sequentially validating each methodological contribution on five downstream development datasets before testing the final configuration on eight untouched test datasets. To reduce the search space and disentangle the effects of SSL pre-training from other basic design choices, we choose to keep some parameters fixed based on best practices in the domain:
(**i**) The architecture used is always the same state-of-the-art residual encoder U-Net architecture (ResEnc U-Net) [21, 38]. (**ii**) The input patch size is set to [160x160x160]. (**iii**) All images are resampled to the same target spacing of [1x1x1] mm [31]. (**iv**) All images are z-score normalized to zero mean and unit variance [20]. (**v**) As optimizer, we use SGD with a decreasing poly-learning rate [5] following nnU-Net [20]. (**vi**) We always employ random sampling during pre-training, irrespective of the prevalence of the different MR modalities.

Pre-training Dataset To develop our pre-training method, we utilize a proprietary brain MRI dataset sourced from over 44 centers containing over 9k patients comprising a total of approx. 44k 3D MRI scans. Due to the variety of data sources, this dataset contains images from more than 10 different MR scanners, various MR modalities and a diverse patient population. For more details on the distributions, we refer to Figure 3.
Since this data is sourced directly from clinical examinations, it includes empty or broken images, poor-quality images and so-called scout scans used to determine the field of view of the patient in the MR. Since these scans are not used in diagnostics, these images are filtered by discarding (a) images with a field of view of $< 50mm$ in any axis, (b) images with a spacing $> 6.5mm$ in any direction and (c) images of file size $< 200kb$, which indicate an empty image. Moreover due to the low quantity of MR Angiography, Susceptibility weighted Images (SWI) and Proton Density (PD) weighted images, we restrict our training data to only include T1, T2, T1 FLAIR and T2 FLAIR images, resulting in our final pre-training dataset of 39,168 MR images.

Development Datasets After pre-training we fine-tune on five datasets and calculate the average Dice Similarity Coefficient (DSC) per dataset to evaluate the effectiveness of the pre-training. The multiple datasets are essential to ensure that our design choices do not overfit to a specific MRI modality or pathological target. Specifically, we utilize:

1. *MS FLAIR (D1)*: Consensus delineations of multiple sclerosis (MS) lesions on T2-weighted FLAIR images [26].
2. *Brain Mets (D2)*: Brain metastases imaged through T1, contrast enhanced gradient echo T1ce, contrast enhanced spin echo T1 and a T2 FLAIR sequence acquired in the Stanford University Hospital [12].
3. *Hippocampus (D3)*: The Hippocampus dataset, task 4 of the medical segmentation decathlon (MSD) [1], contains delineations of the anterior and posterior Hippocampus in T1 weighted MRI [32].
4. *Atlas22 (D4)*: Anatomical Tracings of Lesions After Stroke (ATLAS) on T1 weighted images. We use the Atlas R2.0 dataset from Liew et al. [24].
5. *CrossModa (D5)*: Delineations of intra-meatal and extra-meatal vestibular schwannoma tumors and cochlea delineations in contrast-enhanced T1 weighted MRI [8].

Of all these datasets, we set aside a hold-out test set comprising 20% of all images before the start of the method development process. The remaining images were split 80/20 into training and validation sets for the development process.

Test Datasets Additionally, eight hold-out test datasets were used to evaluate the efficacy of our learned representations when fine-tuning them to segment other target structures.

1. *Cosmos (D6):* This dataset features carotid vessel wall segmentation and atherosclerosis diagnosis, of which we use the contours to evaluate segmentation performance [4].
2. *HaNSeg (D7):* This dataset contains segmentations of 30 organs at risk (OAR), with associated T1 MRI and CT, of which we only use the MR images for model development [29].
3. *Isles22 (D8):* This dataset contains annotations of ischemic stroke lesions, with associated diffusion-weighted imaging (DWI), apparent diffusion coefficient (ADC) and T2 FLAIR [18].
4. *HNTS-MRG24 (D9):* T2-weighted MR images of pre-treatment oropharyngeal cancer and metastatic lymph nodes with associated annotations [42].
5. *BraTS Africa (D10):* This dataset contains 1.5 Tesla T1, T1ce, T2, and T2 FLAIR MRIs of glioblastoma and high-grade gliomas imaged in Nigeria.
6. *T2 Aneurysms (D11):* This proprietary dataset contains 240 T2 MRI images of segmented brain aneurysms with the surrounding brain tissue.
7. *TOF Angiography Aneurysms (D12):* This proprietary dataset contains 144 time-of-flight MR-angiography im-

ages of segmented brain aneurysms with the surrounding brain tissue.

8. *BraTS Mets (D13):* This dataset holds brain metastases segmentations on T1ce MRIs, similar to *D2*. However, instead of fine-tuning on it we use it to measure generalization when inferring models trained on *D2* on it.

For test datasets *D6-D12* we use an 80/20 split for fine-tuning and for testing. A validation set is omitted as we fine-tune each method only once without any interventions. For *D13* all data is used for testing, as no training is conducted. *D12* and *D13* are only used to assess generalization performance, with *D13* only appearing in Appendix Tab. 9.

3. Related Work

In this section, we revisit currently established MAE pre-training paradigms. Then, we survey existing MAE pre-training paradigms in the 3D medical domain, before expanding this to a broader set of pre-training paradigms for 3D medical image segmentation.

3.1. Masked Autoencoders

Auto-encoders have been established for a long time through the early pioneering works of Vincent et al. [41] but recently garnered significant attention through He et al. [16], showing that a simple MAE pre-training paradigm is scalable and provides substantial performance improvements. Subsequently, follow-up works extended these findings to video data [10, 37] as well as language-image data [23]. These earlier successes primarily relied on Transformer architectures, as their sequence modeling offered significant VRAM savings. CNNs on the other hand are less suited to such a pre-training paradigm as masking disrupts the 2D data-structure they compute on. Despite this Woo et al. [45] and Tian et al. [35] were able to extend MAE pre-training successfully to CNNs. Woo et al. [45] introduced their fully convolutional masked autoencoder (FC-MAE) framework, while Tian et al. [35] introduced adaptations to the convolution computation and normalization to make it resilient to masking.

3.2. Masked Autoencoder In 3D Medical

The earliest work using autoencoders for self-supervised pre-training in the medical domain was Models Genesis (MG) conducted by Zhou et al. [48] in 2019. They introduced reconstruction of in-painting, out-painting, and denoising of local intensity shifts as their pre-training task. Many approaches followed, with Swin UNETR by Tang et al. [34] proposing a joint in-painting, rotation, and contrastive objective for Swin Transformer pre-training. Chen et al. [6] proposed a basic masking and reconstruction strategy applied to 3D Vision Transformers (ViT) [9] termed (3DMAE) and Zhuang et al. [49] proposed to use a combination of masked global and local views in conjunction with a reconstruction and consistency loss to pre-train 3D ViTs, named GL-MAE. MAPSeg by Zhang et al. [47] combines a masked autoencoder objective and pseudo-labeling to address domain shifts in a federated setting. More recently, in concurrent work, Munk et al. [25] proposed pre-training CNNs with a vanilla masking and reconstruction objective without any adaptations on a large brain dataset (AMAES) and Tang et al. [33] presented a sparse MAE with reconstruction objective pre-trained on abdomen CT data (HySparK).

3.3. Other Pre-Training For 3D Medical

Additionally, self-supervised learning approaches for 3D medical images include Zhuang et al. [50], who introduced a training objective aimed at recovering the orientation and rotation of shuffled and rotated sub-volumes, inspired by a Rubik's cube. Wang et al. [43] proposed a method that involves mixing two images at varying patch rates and predicting the mixing categories via a segmentation objective, termed Volume Fusion (VF). Geometric Visual Similarity Learning (GVSL) of He et al. [17] leveraged anatomical consistency by learning to register two images of the same field of view and using local similarity as a training objective. More recently, Wu et al. [46] suggested partitioning a volume into non-overlapping base-crops and generating additional overlapping sub-crops, with a training objective to predict the degree of overlap between sub-crops and base-crops (VoCo).

4. Revisiting 3D MAEs

Masked autoencoders (MAEs) are a well-established pre-training paradigm in the natural imaging domain and in the medical image segmentation domain for transformers. In this section, we investigate this paradigm and optimize it for 3D medical image segmentation using a ResEnc U-Net architecture of Isensee et al. [21].

Default parameters MAEs are trained by masking an input image to a certain degree and training the network to reconstruct the occluded regions, minimizing deviations between reconstruction and the original image. In our experiments, we train the MAE with an L2-Loss in the z-score normalized voxel space and only calculate the reconstruction loss where regions were masked. Moreover, we do not remove skip-connections, following the general consensus of Woo et al. [45], Tian et al. [35], and He et al. [16]. The default hyperparameters (as used by the model denoted in gray in Tab. 2) are learning rate 1e-2, weight decay 3e-5, batch size 6, SGD optimizer with Nesterov momentum 0.99, masking ratio of 75% trained with a PolyLR schedule for 250k steps (this represent 1000 epochs in the nnU-Net framework) and minor spatial augmentations of affine scaling, rotation and mirroring.

Table 2. **Development Experiments:** During development, we evaluate the Average DSC on all development datasets to quantify the best configuration. Methods of the same configuration are denoted through common colors.

(a) **Sparsification:** The introduction of all sparsification components showed the best results, with most improvements coming from adding the additional densification conv layer between encoder and decoder.

Configuration	D1	D2	D3	D4	D5	Avg. D1-D5
Base	49.96	72.62	**89.03**	**63.45**	81.67	71.35
+ Sparse Conv. + BN.	50.34	73.34	88.91	62.96	81.25	71.36
+ w/ Mask Token	50.02	73.21	88.92	62.84	**81.83**	71.37
+ w/ Dens. Conv.	**51.02**	**74.07**	88.91	62.81	81.50	**71.66**

(b) **Masking ratio:** Masking ratios between 60% and 75% worked best when choosing static ratios. However, a dynamic range, including higher masking ratios, performs similarly and was chosen.

Mask ratio	D1	D2	D3	D4	D5	Avg. D1-D5
30 %	50.25	71.37	88.90	63.18	81.70	71.08
45 %	50.60	70.83	88.97	63.27	**81.73**	71.08
60 %	50.62	73.56	88.96	**63.37**	81.48	71.60
75 %	51.02	**74.07**	88.91	62.81	81.50	**71.66**
90 %	50.56	72.51	**88.98**	62.41	81.49	71.19
U[60%-90%]	**51.49**	74.01	88.83	62.39	81.54	71.65

(c) **Development performance:** When comparing our final S3D model against the baseline methods, trained equally, S3D exceeds all baselines on the development datasets.

Pre-training	D1	D2	D3	D4	D5	Avg. D1-D5
No Dyn.	45.56	72.26	88.80	60.44	**82.61**	69.93
No Fixed	49.37	69.13	88.78	60.74	81.33	69.87
VoCo	50.35	67.20	88.22	57.82	80.29	68.77
VF	49.93	69.58	**88.83**	61.75	81.48	70.31
MG	50.50	71.14	**88.83**	**63.29**	82.15	71.18
S3D (ours)	**51.49**	**74.01**	**88.83**	62.39	81.54	**71.65**

Sparsification When masking the input image, CNNs are not able to ignore the masked regions in the same manner as transformers can. To address this, Tian et al. [35] proposed to adapt the CNN architectures to better fit the sparse inputs: **(a) Sparse Convolutions and Normalization:** Through the receptive field of convolutions masked-out regions are iteratively eroded from their boundaries. By re-applying the masked regions after every convolution this problem can be resolved. Moreover, the masks can introduce a problematic shift in the normalization layer statistics due to the introduced zero values. To resolve this normalization is constrained to only consider the non-masked values. **(b) Mask Token:** Instead of feeding the light-weight decoder feature maps with zeroed mask regions, the regions are densified by filling them with a learnable mask token, simplifying the reconstruction task of the decoder. **(c) Densification Convolution:** After filling the masks with the Mask Token and before passing the feature maps to the decoder, a [3x3x3] convolution is applied to the feature maps at every resolution except the highest resolution to prepare the representations for decoding.

Results of these changes are visualized in Tab. 2a. The adaptations are introduced iteratively, meaning the 'Mask-Token' ablation is only applied together with the Sparse Convolutions and Normalization. It can be observed that the full set of adaptations improves performance by an average of 0.3 DSC points across our development datasets. Subsequently, all adaptations are kept and the following evaluations are presented with these changes applied.

Masking strategy The masked region is determined by sampling randomly in the CNN's bottleneck of shape [5x5x5] and up-sampling these regions to the input resolution, to ensure the masks align in the bottleneck of the CNN architectures. This results in masking regions of [32x32x32] voxels of non-overlapping regions in the input. As a sampling strategy, we follow random masking, as previous work showed no benefit of structured masking for images nor videos [10, 16]. In the scope of the development phase, we explore 5 static masking ratios between 30% and 90% and evaluate a dynamic masking ratio randomly masking between 60% and 90%.

Results are presented in Tab. 2b and highlight that the masking ratios of 60%, 75%, and the dynamic masking ratio of 60% to 90% perform equally well. Due to the highly similar performance, we choose to proceed with a dynamic masking ratio over the static masking ratio, due to expecting this masking to be more difficult to learn. We refer to this model as Spark3D (S3D) going forward.

Fine-tuning strategy Given a pre-trained model, a crucial question arises: Which weights to transfer and how to schedule the fine-tuning? We investigated various different schedules, with all of them implemented in the nnU-Net framework [20]. Regarding weight transfer we investigated transferring (i) both the encoder and decoder, or (ii) only the encoder with a randomly initialized decoder. Regarding the fine-tuning schedule, we investigate whether to use a learning rate warm-up of 12.5k steps, ramping the learning rate up to the maximum LR. When only transferring the encoder, an additional warm-up of only the decoder is investigated to adapt the randomly initialized decoder to the pre-trained encoder. In some configurations, this results in two learning rate warm-ups of 12.5k steps each. Additionally, we investigate whether to keep the encoder frozen for the entire fine-tuning process or to fine-tune the encoder weights as well. Lastly, we investigate whether to decrease the peak learning rate to 1e-3, 1e-4 or to keep it at the default of 1e-2.

Results are presented in Table 3 and allow three important observations to be made: (i) **Warm-up stages are essential:** Not applying a warm-up step significantly reduces performance. Including a warm-up for both the encoder and decoder boosts accuracy by 0.6 to 1 DSC points. (ii) **Learning rate adjustments matter:** Reducing the peak learning

Table 3. **Fine-tuning matters.** We compare various combinations of weight transfer and fine-tuning schedules. Transfer: Transfer all weights, Transfer encoder weights only. Warm-Up and Fine-tuning: Only decoder weights adapted during fine-tuning, Encoder and decoder weights adapted during fine-tuning. ‡: nnU-Net default (Dynamic planning.)

Transfer	1. Warm-Up	2. Warm-Up	Fine-tuning	Max. LR	D1	D2	D3	D4	D5	Avg
	-	-		1e-2	45.56	72.26	88.80	60.44	82.61	69.93‡
	-	-		1e-2	49.37	69.13	88.78	60.74	81.33	69.87
	-	-		1e-2	50.37	70.64	88.61	61.51	81.91	70.61
	-	-		1e-3	49.98	71.04	88.68	61.45	82.12	70.65
	-			1e-2	49.84	72.56	88.45	62.16	81.75	70.95
	-			1e-3	51.54	72.74	88.85	62.44	82.33	71.58
	-			1e-4	50.66	72.98	88.68	62.73	82.09	71.43
	-	-		1e-2	50.81	68.55	88.52	60.73	80.77	69.87
	-	-		1e-3	50.11	73.90	88.58	61.82	81.48	71.18
	-			1e-2	51.39	72.06	88.65	61.70	81.78	71.12
	-			1e-3	51.04	72.72	88.91	62.84	81.19	71.34
				1e-2	48.81	72.92	88.84	62.46	82.04	71.02
				1e-3	51.42	72.84	89.09	63.30	82.15	**71.76**
				1e-4	50.13	72.26	88.72	61.70	81.93	70.95

rate to 1e-3 during fine-tuning consistently yields better results than the default 1e-2, with the best performance seen when fine-tuning both the encoder and decoder with lower learning rates. (iii) **Freezing encoder weights is detrimental:** The encoder should not remain fixed during fine-tuning. Allowing the encoder to be fine-tuned improves the model's performance compared to when only the decoder is fine-tuned.

5. Results and Discussion

We compare our final model S3D against Volume Contrastive (VoCo) [46], VolumeFusion (VF) [43], as well as Models Genesis (MG) [48]. The baselines are pre-trained for 250k steps using the same framework on the same data with the same backbone and the same hyperparameters (where possible) but scaled to fully utilize an A100 40GB GPU. We provide explicit baseline methods and configuration details in Appendix A. Moreover, we compare against two strong from-scratch baselines. The first, *No (Dyn.)*, represents a from scratch default nnU-Net 3D architecture that was planned and preprocessed on each downstream dataset individually, potentially resulting in different architectures, data preprocessing, and spacings. The second from-scratch baseline, *No (Fixed)*, is a nnU-Net training with the same plans and pre-processing as defined by our pre-training. The dataset-wise mean dice similarity coefficient (DSC) and mean normalized surface distance (NSD) at 2 mm tolerance values across our test dataset suite are provided in Tab. 4. Additionally, the ranking stability is evaluated through bootstrapping and is provided in Fig. 2.

5.1. Observations

SSL pre-training works Across all tested datasets, SSL pre-trained methods demonstrate improved downstream segmentation performance. Comparing our S3D method to the most similar from-scratch baseline *No (Fixed)*, we observe higher DSC scores in 10 out of 11 test datasets, with an average increase of +2 DSC points and +1.6 NSD points. This improvement is not limited to our method; MG and VF also achieve higher performance than the baseline, indicating the utility of SSL methods when applied to sufficient data and a state-of-the-art architecture.

MAEs dominate Throughout our test dataset pool, SSL schemes using the masked image modeling paradigm (MG and S3D) consistently rank higher than the contrastive VoCo or the pseudo-segmentation-based VolumeFusion pre-training method for CNN pre-training. Given the age of Models Genesis - published in 2019 - it is surprising to see it outperform the more recent VoCo or VF. We attribute this to a combination of two factors: 1. Models Genesis was originally published and trained on an outdated 3D-UNet [7] and outside of the powerful nnU-Net framework [20]. This highlights the importance of avoiding Pitfall 2: Training on a state-of-the-art backbone. 2. VoCo and VF were introduced in conjunction with architectures they were optimized for. By transferring them to a CNN setting, hyperparameters chosen to optimize the method for their original architecture pre-training combination may be suboptimal for the new CNN backbone.

Table 4. **S3D out-performs all baselines:** The upper table presents the DSC results, while the lower table displays the NSD results across the test datasets, covering a diverse range of brain MRI tasks. *No Fixed* represents a from-scratch baseline sharing the same architecture, preprocessing and downstream training steps as all SSL methods. *No Dyn.* represents the original nnU-Net adapted to each downstream dataset individually.

SSL Method	No (Dyn.)	No (Fix.)	VoCo	VF	MG	**S3D**
Dataset	**Dice Similarity Coefficient (DSC)**					
MS FLAIR (D1)	57.81	59.82	59.70	59.29	58.64	**60.35**
Brain Mets (D2)	63.66	56.53	56.25	61.01	**65.39**	65.24
Hippocampus (D3)	89.18	89.24	88.78	89.03	89.38	**89.60**
Atlas22 (D4)	63.28	65.52	62.97	65.76	65.93	**66.95**
CrossModa (D5)	**85.64**	83.44	83.07	84.24	83.91	84.08
Cosmos22 (D6)	60.28	78.17	77.40	**80.09**	79.67	80.00
ISLES22 (D7)	77.94	79.44	78.14	78.96	78.85	**79.70**
Hanseg (D8)	59.00	61.85	57.47	61.49	**62.52**	62.11
HNTS-MRG24 (D9)	66.73	65.90	67.65	63.34	68.00	**68.62**
BRATS24 Africa (D10)	**93.07**	92.51	91.97	92.16	92.36	92.19
T2 Aneurysms (D11)	46.76	41.97	40.16	44.96	45.48	**47.26**
Avg. DSC	69.40	70.40	69.41	70.94	71.83	**72.37**
Avg. Rank	4.64	4.55	6.27	4.36	3.18	**2.00**
Dataset	**Normalized Surface Distance (NSD)**					
MS FLAIR (D1)	78.77	**80.16**	79.70	79.57	79.16	80.03
Brain Mets (D2)	80.72	76.72	72.77	79.20	81.51	**82.53**
Hippocampus (D3)	**99.46**	99.42	99.43	99.46	99.39	99.46
Atlas22 (D4)	70.52	73.77	70.15	73.67	74.22	**75.35**
CrossModa (D5)	**99.85**	99.76	99.72	99.78	99.74	99.81
Cosmos22 (D6)	72.60	96.47	94.48	96.89	96.95	**97.45**
ISLES22 (D7)	88.55	90.45	89.39	90.28	89.59	**90.59**
Hanseg (D8)	82.20	**85.94**	80.44	85.29	85.94	85.80
HNTS-MRG24 (D9)	71.83	71.26	73.47	67.88	73.22	**74.07**
BRATS24 Africa (D10)	**95.66**	95.36	94.95	94.94	95.33	95.06
T2 Aneurysms (D11)	**62.24**	55.56	51.79	58.97	59.38	61.18
Avg. NSD	82.04	84.08	82.39	84.17	84.95	**85.58**
Avg. NSD Rank	4.27	4.27	5.82	4.64	4.00	**2.18**

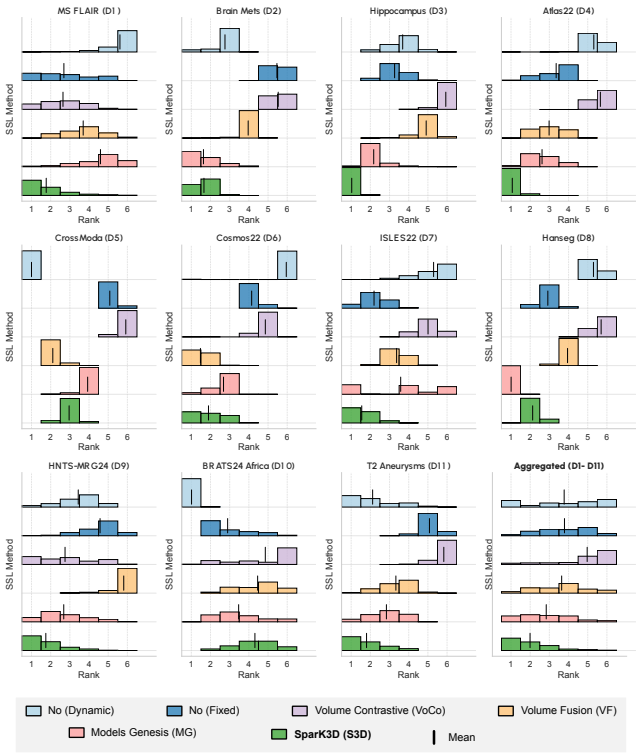

Figure 2. **S3D ranks best across all methods.** In addition to absolute mean performance, we report the ranking stability of the methods through bootstrapping for all test datasets as well as the aggregated rank across all datasets.

Impact of dynamic configuration Comparing the *No (Dyn.)* and *No (Fixed)* configurations, both trained from scratch, reveals that selecting the appropriate configuration for each dataset can significantly influence performance. For instance, on datasets D2 and D11, the dynamic configuration outperforms the fixed by +7 and +5 DSC points, respectively, while for D6, the fixed configuration yields results +18 DSC points higher. In the majority of datasets where the fixed configuration underperforms relative to the dynamic nnU-Net, pre-training helps to recover performance. However, in some cases, such as with D5, the dynamic default nnU-Net still proves superior.

5.2. Ablation experiments

Low-Data Regime While previous experiments focused on comparing SSL pre-training against a from-scratch baseline with full-scale datasets available, many applications in the medical domain have access to only a very small amount of labeled images. To measure the benefits of pre-training in such a low-data regime, we artificially reduced the total amount of data available for training to 10, 20, 30, or 40 labeled images.

Results are presented in Tab. 6. Our pre-trained S3D model leads to better downstream performance compared to training from scratch in this setting. With just 40 trained images, the fine-tuned model nearly matches the performance of the from-scratch model trained on the full dataset. Future research could explore whether optimizing the training duration or learning rate schedule could prevent the pre-trained network from overfitting during fine-tuning on such a limited number of training images.

Generalization Performance To assess the generalization capability of the proposed pre-training method, we tested two scenarios. First, we evaluated fine-tuning our method on an unseen modality using the TOF Angiography Aneurysms dataset (D12). As shown in Tab. 9 without pre-training, the fixed configuration suffers a performance drop of 20 Dice points. We attribute this to the significant difference in median spacing for the downstream task ($[0.50, 0.43, 0.43]$ mm), which has a higher resolution than the fixed target spacing of $[1, 1, 1]$ mm used in the pre-training experiments. This lower resolution likely increases the difficulty of segmenting small aneurysms. Despite this decrease, pre-training mitigates some of this degradation and proves highly beneficial compared to training the same

configuration from scratch. Interestingly, Models Genesis achieves the best results, potentially due to its use of intensity augmentations during pre-training, which increases robustness against brightness shifts, such as when generalizing to different MRI sequences. Second, we fine-tuned on the D2 dataset using only the T1 contrast-enhanced (T1ce) sequence and applied these models directly to D13 without any additional fine-tuning. While the dynamic configuration performed best on the in-distribution validation cases of D2, the results on D13 indicate that pre-training improves generalization across different centers, with our S3D model yielding the best performance.

Pre-training Time Previous studies have demonstrated the positive impact of extended training schedules on the quality of learned representations for downstream tasks [10, 16]. To explore the relevance of this factor in the 3D medical domain, we conducted a similar experiment, evaluating training durations ranging from 62.5k to 1M steps. Our results indicate that the benefits of longer training schedules begin to degrade after 250k steps, as evident in Tab. 7.

Fine-tuning length Initializing from pre-trained weights has the potential to reduce the computational resources needed for the network to adapt to new tasks. To assess this, we tested different fine-tuning durations on our development datasets. While maintaining 12.5k iterations for both the decoder and full network warm-up phases, we experimented with varying subsequent training lengths. As shown in Tab. 10, adding just 12.5k additional iterations (37.5k total) already outperforms training from scratch. However, achieving optimal performance still requires completing the full fine-tuning schedule.

Multi-Channel Input In many medical examinations, it is common to perform multiple scans, as clinicians often require images with different characteristics for accurate decision-making. Consequently, some datasets, like D2, D8, and D10, contain multiple input modalities. While pre-training may involve all modalities, we only feed one modality at a time into the network since not all patients have scans in every modality. This raises the question of how to handle datasets with multiple registered images. To address this, we conducted a 5-fold cross-validation on the D2 development dataset. We evaluated the replication of each input modality along with random initialization of the input stem weights. Additionally, we tested freezing the stem weights during the decoder warm-up phase. As shown in Tab. 8, the most stable and consistently effective approach was replicating the pre-trained stem and keeping it frozen during the decoder's warm-up period.

6. Conclusion

This work is the first to demonstrate the potential of properly configured MAEs in 3D medical image segmentation. By overcoming key pitfalls in previous research, such as limited dataset sizes, outdated architectures, and insufficient evaluation, we show a consistent performance improvement over previous SSL methods. Notably, for the first time, we achieve consistent improvements over the dynamic, dataset-adaptive nnU-Net baseline, validated across a large and diverse set of development and testing datasets.

Limitations In our current manuscript, we justify our selection of baseline methods by excluding prior work that was specific to architectures (3DMAE), methods that had no public code (GL-MAE), and methods that do not outperform a nnU-Net baseline trained from scratch (see Appendix Tab. 11). While our findings are promising, several avenues remain open for future exploration. We limited our method to Head and Neck MRI images, allowing us to focus on this region with a comparatively large pretraining dataset. Expanding to encompass the entire body across multiple imaging modalities would have required substantially more data to achieve comparable performance. Notably, in our ablations, we tested increasing training time and batch size (Appendix Tab. 12) which did not lead to performance improvements, but the question remains of whether scaling the pre-training dataset size or the model parameters could enable further improvements. Furthermore, the intensity shifts employed by Model Genesis SSL task hint at intriguing possibilities for improving generalization across unseen MRI modalities, which needs to be further explored for MAEs. Lastly, a data-centric approach to curating the most relevant data for SSL represents an exciting frontier for future research. Raw clinical datasets often contain images not intended for diagnostic purposes, such as those used for scanner calibration, which can dilute the effectiveness of pre-training. While we applied basic filtering to exclude low-quality data, more sophisticated filtering techniques could significantly enhance the quality of the pre-training process.

To conclude, this work follows the spirit of prior studies like nnU-Net [20] showing that a robust development strategy, informed model configuration, and rigorous validation leads to true and sustainable performance improvements, contrasting the current hype for employing and modifying the latest network architecture. With this paper, we hope to contribute to a cultural shift in the community toward validation-driven development enabling true scientific progress.

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